

HDOC Day-27

Generation of Pyramid and Diamond Patterns Using a Menu-Driven C Program

Aim

To write a menu-driven C program using a switch statement and nested loops to generate different pyramid and diamond patterns. Each pattern is implemented using a separate parameterless function with no return value, without using arrays.

Algorithm

1. Start the program.
2. Display a menu containing Full Pyramid, Inverted Pyramid, Diamond Pattern, Palindromic Number Pyramid, Repeated-Number Pyramid, and Exit.
3. Accept the user's choice.
4. Use a switch statement to process the selected option.
5. Call the corresponding parameterless function.
6. Inside the selected function, input the number of rows n.
7. Use nested loops to control spaces, rows, asterisks, or numbers as required.
8. Display the selected pattern.
9. If the user enters an invalid choice, display an appropriate error message.
10. Continue displaying the menu until the user selects Exit.
11. Stop the program.

C Program

```
#include <stdio.h>

void fullPyramid(void);
void invertedPyramid(void);
void diamondPattern(void);
void palindromicPyramid(void);
void repeatedNumberPyramid(void);

int main()
{
    int choice;

    do
    {
        printf("\n----- PYRAMID/DIAMOND PATTERNS MENU ----- \n");
        printf("1. Full Pyramid\n");
        printf("2. Inverted Pyramid\n");
        printf("3. Diamond Pattern\n");
        printf("4. Palindromic Number Pyramid\n");
```

```

printf("5. Repeated-Number Pyramid\n");
printf("6. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1: fullPyramid(); break;
    case 2: invertedPyramid(); break;
    case 3: diamondPattern(); break;
    case 4: palindromicPyramid(); break;
    case 5: repeatedNumberPyramid(); break;
    case 6: printf("Exiting program...\n"); break;
    default: printf("Invalid choice! Please enter a number from 1 to 6.\n");
}
} while (choice != 6);
return 0;
}

void fullPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: "); scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("* ");
        printf("\n");
    }
}

void invertedPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: "); scanf("%d", &n);
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("* ");
        printf("\n");
    }
}

void diamondPattern(void)
{
    int n, i, j;
    printf("Enter number of rows for upper half: "); scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("* ");
        printf("\n");
    }
    for (i = n - 1; i >= 1; i--) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("* ");
        printf("\n");
    }
}

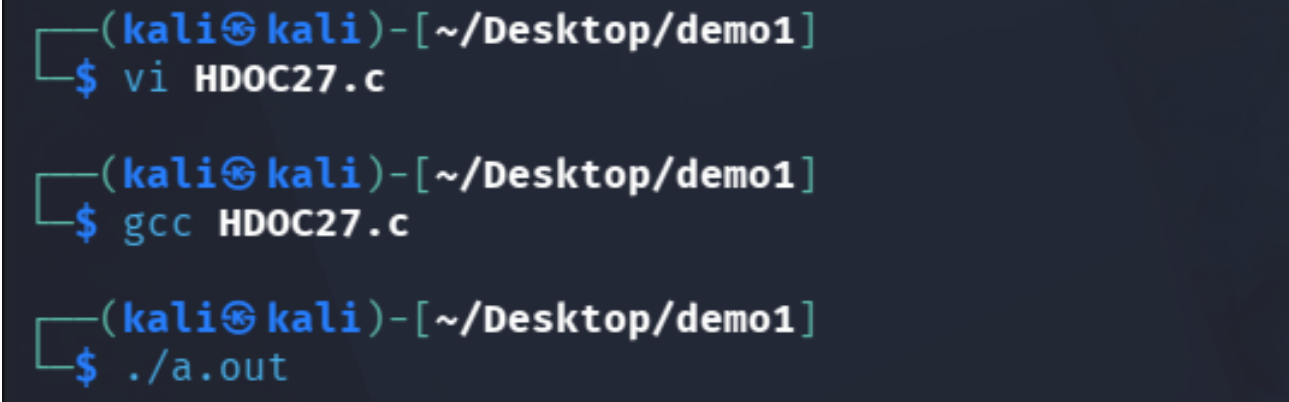
void palindromicPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: "); scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("%d ", j);
        for (j = i - 1; j >= 1; j--) printf("%d ", j);
        printf("\n");
    }
}

void repeatedNumberPyramid(void)

```

```
{
    int n, i, j;
    printf("Enter number of rows: "); scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= 2 * i - 1; j++) printf("%d ", i);
        printf("\n");
    }
}
```

Compilation/Execution Evidence



```
(kali㉿kali)-[~/Desktop/demo1]
└─$ vi HD0C27.c

(kali㉿kali)-[~/Desktop/demo1]
└─$ gcc HD0C27.c

(kali㉿kali)-[~/Desktop/demo1]
└─$ ./a.out
```

The image shows a terminal window with three commands executed in sequence. Each command is preceded by a prompt indicating the user is 'kali' on a 'kali' machine in the directory '~/Desktop/demo1'. The first command is 'vi HD0C27.c', the second is 'gcc HD0C27.c', and the third is './a.out'.

—— PYRAMID/DIAMOND PATTERNS MENU ——

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 1

Enter number of rows: 4

```
  *
 * * *
* * * * *
```

—— PYRAMID/DIAMOND PATTERNS MENU ——

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 2

Enter number of rows: 4

```
* * * * *
 * * * *
  * * *
   *
```

—— PYRAMID/DIAMOND PATTERNS MENU ——

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 3

Enter number of rows for upper half: 4

```
  *
 * * *
* * * * *
```

—— PYRAMID/DIAMOND PATTERNS MENU ——

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 4

Enter number of rows: 4

```
    1
  1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

—— PYRAMID/DIAMOND PATTERNS MENU ——

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 5

Enter number of rows: 4

```
    1
  2 2 2
 3 3 3 3 3
4 4 4 4 4 4 4
```

—— PYRAMID/DIAMOND PATTERNS MENU ——

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit

Enter your choice: 7

Invalid choice! Please enter a number from 1 to 6.

```

——— PYRAMID/DIAMOND PATTERNS MENU ———
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 6
Exiting program...

```

Test Case Table

Test Case	Choice	Input	Expected Output
1	1	n = 4	Full pyramid containing 1, 3, 5, 7 asterisks
2	2	n = 4	Inverted pyramid containing 7, 5, 3, 1 asterisks
3	3	n = 4	Diamond with 7 rows and maximum 7 asterisks
4	4	n = 4	Palindromic pyramid ending with 1 2 3 4 3 2 1
5	5	n = 4	Repeated-number pyramid ending with seven 4s
6	7	-	Invalid choice message
7	6	-	Program exits

Result

The menu-driven C program was successfully implemented using switch-case statements, parameterless functions, and nested loops without arrays. It correctly generates the Full Pyramid, Inverted Pyramid, Diamond Pattern, Palindromic Number Pyramid, and Repeated-Number Pyramid. The program also handles invalid menu choices and continues execution until the user selects the Exit option.